

Skills Summary

Interpreting Remainders in Real Situations

Use this reference while completing your worksheet or when studying.

Skill 1 — Counting only the groups that fill up

Rule: Divide, then *drop* the remainder.

Use this when the question says *completely full*, *whole*, or *how many can you make*. A leftover is not enough to make one more group, so it is not counted.

Total $\div$ group size = quotient R remainder $\Rightarrow$ answer = quotient

Example: There are 87 crayons. Each cup holds 6 crayons. How many cups can be filled completely?

Step 1. The question asks for cups that are completely full, so any leftover crayons do not count — this is a drop-the-remainder question.

Step 2. $6 \times 14 = 84$, and $87 - 84 = 3$, so $87 \div 6$ is 14 R 3. The 3 leftover crayons cannot fill a 15th cup. Cups filled completely: **14**

Try it: There are 71 beads. Each bracelet uses 8 beads. How many complete bracelets can be made?

8 : 71 = 8 R 7
check: $8 \times 8 = 64$, $64 + 7 = 71$

Skill 2 — Counting every group you must provide

Rule: Divide, then *add one* whenever there is a remainder.

Use this when the question says *needed*, *must order*, or *so everyone fits*. The leftovers are real things that still need a place, so they need one more group — even a group that is not full.

Example: There are 119 campers. Each canoe holds 6 campers. How many canoes are needed?

Step 1. Every camper must get into a canoe, so leftover campers still need one — this is a round-the-quotient-up question.

Step 2. $6 \times 19 = 114$, and $119 - 114 = 5$, so $119 \div 6$ is 19 R 5. Those 5 campers need a canoe of their own. Canoes needed: **20**

Try it: A van carries 7 people. How many vans are needed for 45 people?

7 : 45 = 6 R 3
check: $6 \times 7 = 42$, $42 + 3 = 45$

Skill 3 — When the leftover is the answer

Rule: The remainder itself is the answer.

Use this when the question says *left over*, *remaining*, or *how much is not used*.

remainder = total - (group size $\times$ quotient)

Example: A florist has 100 roses and makes bunches of 12. How many roses are left after every full bunch is made?

Step 1. The question asks what is not used, so the answer is the remainder itself, not the number of bunches.

Step 2. $12 \times 8 = 96$, and $100 - 96 = 4$, so $100 \div 12$ is 8 R 4. Roses left over: **4**

Try it: There are 62 stickers, put onto sheets of 9. How many stickers are left over?

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Skill 4 — Estimating the quotient first

Rule: Round each number to the nearest ten to get numbers that divide easily, then divide. An estimate tells you roughly how big the answer should be, so a wildly wrong exact answer is easy to spot. An estimate is a check, never the final answer.

Example: Estimate $178 \div 21$ by rounding each number to the nearest ten.

Step 1. Exact division by 21 is awkward, so round both numbers to friendly tens first and divide those instead.

Step 2. 178 rounds to 180 and 21 rounds to 20; then $180 \div 20 = 9$. Estimate: **9**

Try it: Estimate $243 \div 31$ by rounding each number to the nearest ten.

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